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OPEN INNOVATION

AI Artisan

Empowering Indian artists with AI-Powered Design and Marketing Tools

TEAM : CodeCanvas

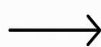
Problem Statement

- India has over 7 million artisans working in handicrafts, weaving, pottery, and dance — yet less than 15% have an online presence.
- Most artisans lack digital literacy, English fluency, and marketing resources, which limits their visibility and income.
- Current e-commerce platforms require professional product images and English content, which are barriers for rural creators.
- High platform fees, poor discoverability, and dependence on intermediaries reduce artisans' profits.
- As a result, traditional art forms are declining, and artisans struggle to sustain their livelihoods.



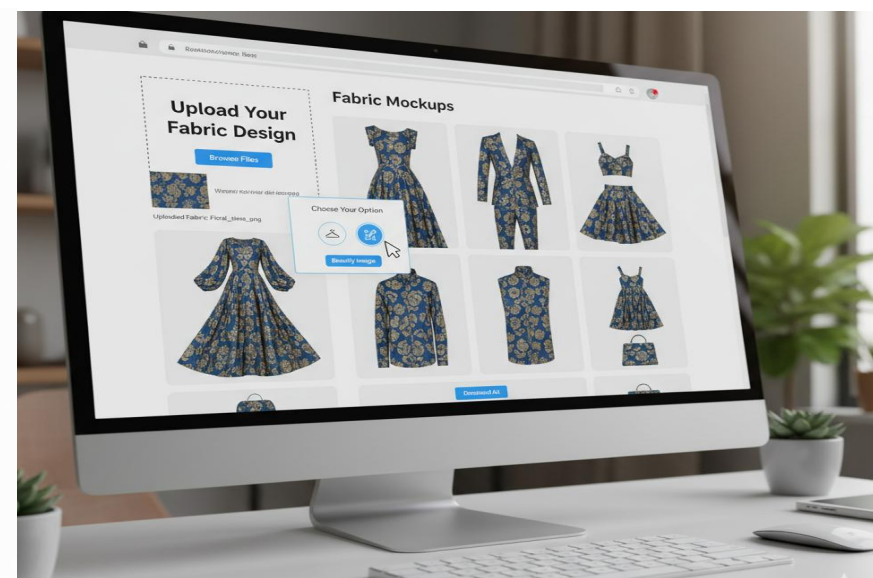
BEFORE

- Artisans can't sell online
- No tech/ English
- Marketing hard



AFTER

- AI Mockups & Filters
- AI Descriptions
- Marketplace
- Multilingual



Solution: AI Artisan Platform

A smart AI-powered marketplace that helps artisans create, promote, and sell their products globally — without needing technical skills.



AI Mockups

- Upload a craft photo → AI auto-generates 3–5 professional mockups on various products (e.g., cushion, saree, poster, jewelry display).



Aesthetic Filters & Enhancements

- One-click professional photo filters (Golden Warm, Pastel, Minimalist, Pencil Art) to make local product photos market-ready.



AI Description Generator

- Automatically writes SEO-optimized, social-media-ready captions and product descriptions in 5+ tones (professional, artistic, emotional, modern, traditional).



Digital marketing training

Simple, text-free videos to show illiterate artisans how to use their phone to grow their business, making digital marketing accessible to all.



Multilingual Interface

- Full translation of product descriptions and UI into regional Indian languages (Telugu, Hindi, Tamil, Bengali) and global languages (English, French, Japanese).



Buy & Sell Marketplace

- Direct platform for listing, selling, and buying products — with secure payment gateway and artisan verification system.



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TECH



Frontend (React)

- **Framework:** React.js (via Vite) for a fast UI
- **Styling:** Tailwind CSS (utility-first design)
- **Animation:** Framer Motion (smooth transitions)
- **Components:** ShadCN UI / Radix UI (accessible, modern)
- **API Calls:** Axios
- **Uploads:** React Dropzone

Backend (Node.js)

- **Framework :** Node.js + Express.js
- **Core Role:** Handles user authentication (JWT), API routing, and database connections.
- **Database ODM:** Mongoose (for MongoDB)
- **File Handling:** Multer (for image uploads)



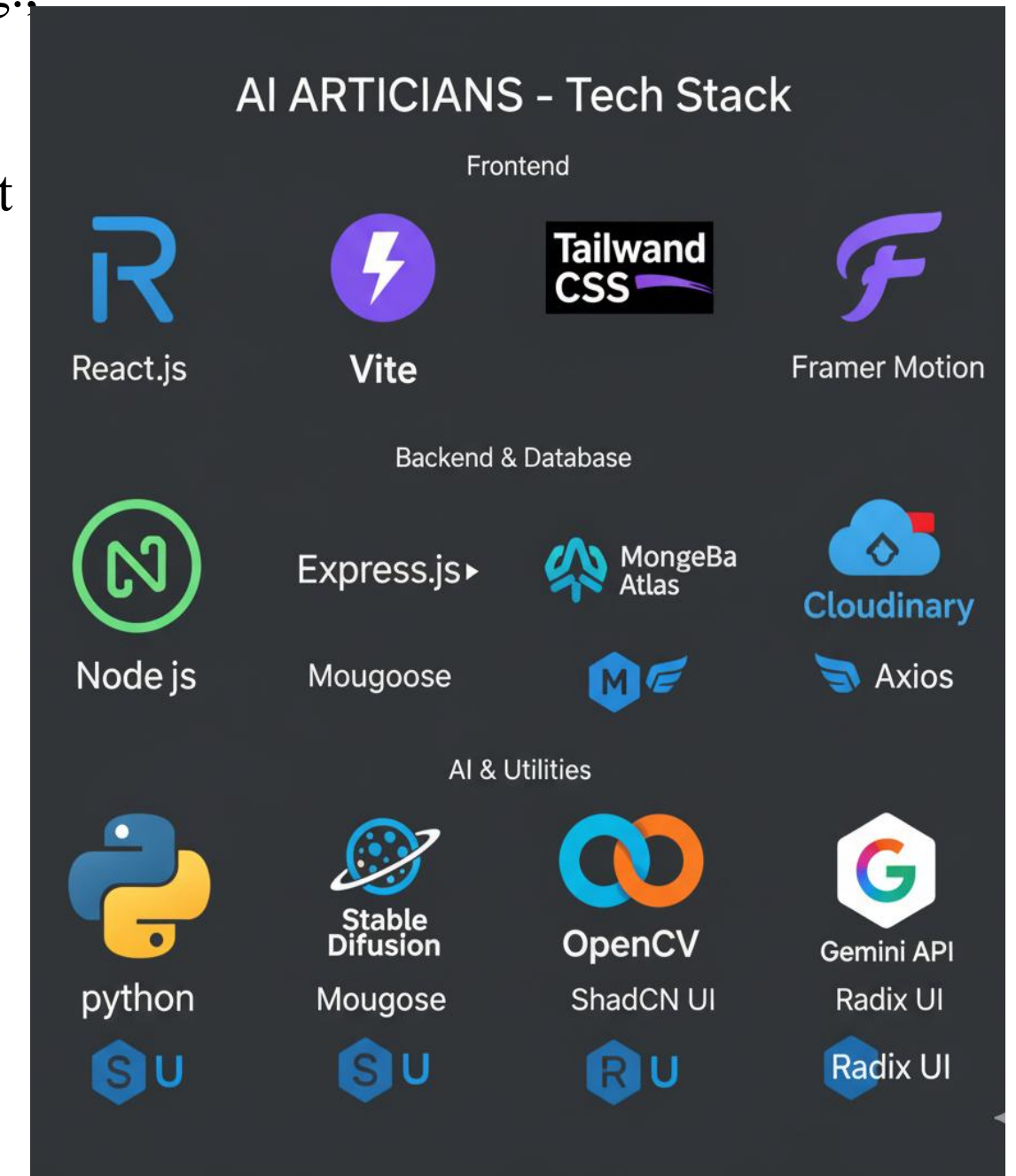
AI Layer (Python)

- **Architecture:** Separate microservice (e.g., FastAPI) for heavy processing.
- **Mockups:** Stable Diffusion + ControlNet
- **Filters:** OpenCV + Pillow
- **Captions:** open-source vision-language model (BLIP)



Database (MongoDB)

- **Service:** MongoDB Atlas (Cloud-hosted NoSQL)
- **Collections :** users, images, mockups, captions
- **Strategy:** Stores metadata and URLs (images stored separately in cloud storage).



BUSINESS SCOPE & IMPACT



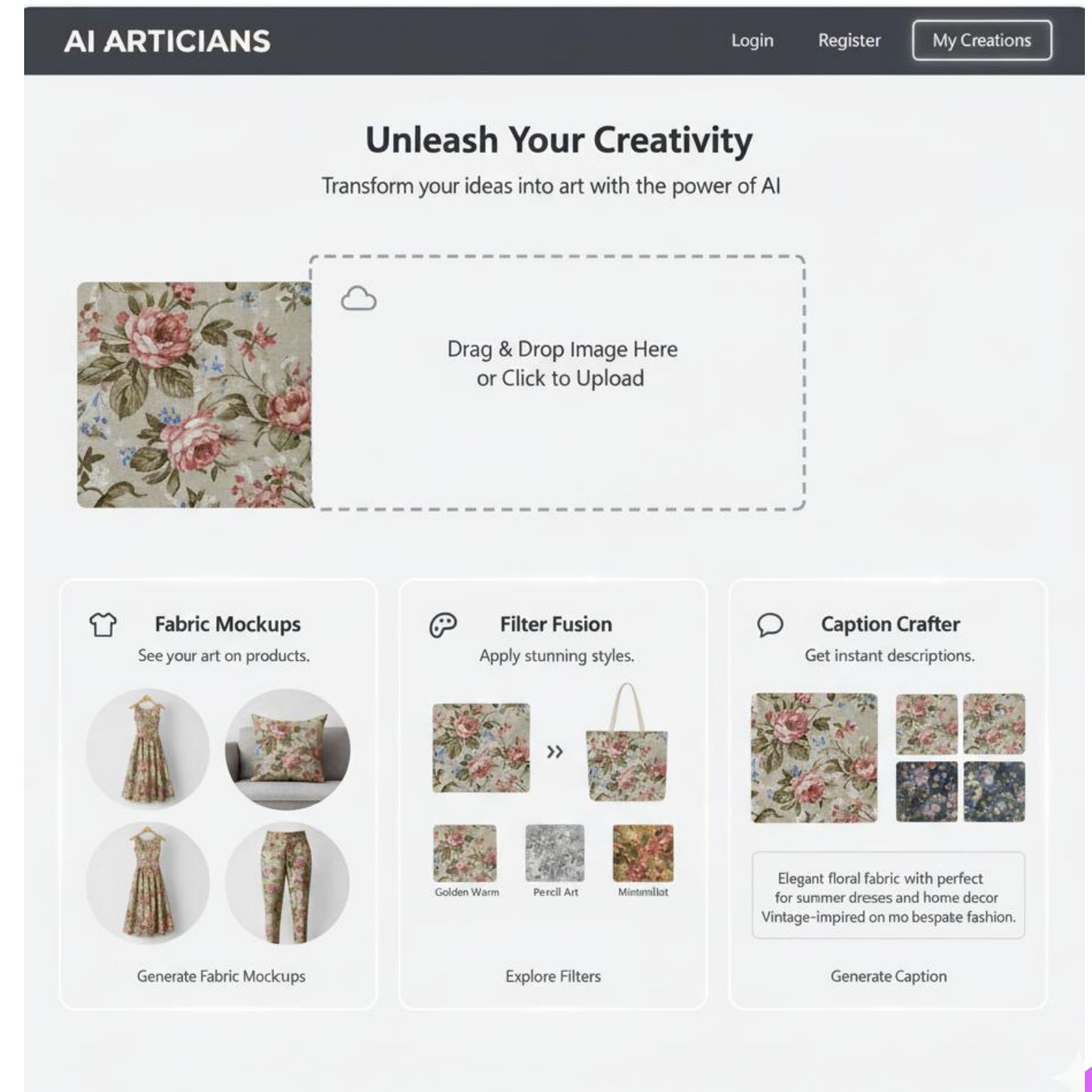
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💰 Revenue Model

- Commission: 5–10% per sale
- Premium Subscription: Access to advanced AI filters and mockup packs
- Partnerships: Collaboration with cultural boards, NGOs, and art fairs
- Corporate Gifting Program: Companies source handcrafted, AI-showcased items

♥ Social & Economic Impact

- Enables digital empowerment for rural artisans
- Encourages women entrepreneurship and sustainable livelihoods
- Preserves Indian cultural heritage through modernization
- Increases visibility and sales for underrepresented artists
- Bridges language and technology barriers using AI and localization



Challenges & Future Scope

⚠️ Current Challenges

- Lack of stable internet access in rural areas
- Convincing artisans to adopt digital tools
- Ensuring AI outputs maintain cultural authenticity
- Handling multilingual data quality and ethical content use

🚀 Future Enhancements

- Voice-based Navigation: For illiterate or semi-literate users.
- AI Recommendation Engine: Suggests trending designs, tags, or colors based on market data.
- Augmented Reality (AR): Buyers can visualize products in real-world environments.
- Blockchain Provenance: Authenticate artisan origin and craft authenticity.
- Global Expansion: Integrate with Etsy, Amazon Handmade, and UNESCO craft initiatives.
- Community Portal: Peer learning, tutorials, and workshops on AI-driven digital marketing.

“AI for culture, not replacement — a bridge between tradition and technology.”

◆ **THANK YOU** ◆



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